

MODE — MULTI-HALO ORBIT DYNAMICS ENGINE

MANY-BODY INTERACTION AND ORGANIZATION FRAMEWORK

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STATUS

Python-based many-body interaction framework focused on bounded organization behavior under tuned balancing dynamics.

The framework is currently intended for:

- nonlinear dynamics research
- complex systems experimentation
- visualization systems
- many-body interaction studies
- simulation architecture development
- GPU / simulation prototyping concepts

TECHNICAL ABSTRACT

This document describes MODE (Multi-Halo Orbit Dynamics Engine), a many-body interaction framework developed in Python for studying emergent large-scale organization behavior under tuned balancing interactions.

The framework investigates how attraction-like, repulsion-like, damping, and range-limited interaction structures can produce persistent organized states including:

- core-halo organization
- orbit-like motion
- collision reorganization
- trajectory curvature behavior
- binary inspiral dynamics
- compression/rebound structures
- moving-center tracking
- nonlinear multi-halo coexistence

The system uses deterministic update rules and controlled state evolution to maintain repeatable simulation behavior under fixed initial conditions.

A non-singular interaction structure was introduced to reduce instability during high-density compression events, allowing the framework to transition between inward compression, rebound, reorganization, and re-expansion regimes without immediate numerical divergence.

Across multiple simulation branches, the framework demonstrated repeatable synthetic behaviors including:

- stable core-halo organization
- multi-halo coexistence
- moving-center tracking
- orbit-like motion
- collision survival and reorganization
- perihelion-style precession response under modified corrections
- path-curvature and lensing-style behavior
- binary inspiral-style organization
- breathing-mode collapse/re-expansion dynamics
- partial coordinated scaling behavior

The framework does NOT rely on Newtonian gravity equations or General Relativity field equations as its primary interaction model.

Instead, the framework investigates whether gravity-like organizational behavior can emerge from tuned balancing interactions between many simultaneous interacting bodies.

Scaling experiments showed that naive geometric rescaling alone did not preserve structure; however, coordinated rescaling of spacing, interaction range, softening parameters, velocity scale, and timestep produced partially preserved halo-ratio behavior, suggesting possible approximate dimensionless similarity within limited operating regions.

The framework is currently best understood as a nonlinear many-body organization and interaction platform suitable for further study in:

- complex systems research
 - visualization systems
 - swarm dynamics
 - educational simulation
 - GPU-accelerated experimentation
 - procedural simulation systems
 - organization dynamics research
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CORE INTERACTION STRUCTURE

MODE combines multiple balancing interaction mechanisms including:

- attraction-like interactions
- short-range repulsive stabilization
- velocity-dependent repulsion
- damping terms
- long-range dispersive pressure
- bounded force limiting
- coordinated parameter balancing

The strongest repeated result observed throughout testing was:

attraction alone -> collapse
repulsion alone -> dispersion
balanced interactions -> persistent organization

This repeatedly produced:

- stable core-halo structures
- orbit-like behavior
- long-lived organization

- moving-center tracking
 - collision resilience
 - rebound/reorganization dynamics
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SELECTED BENCHMARK TESTS

1. CORE-HALO ORGANIZATION

Repeatedly observed:

dense central organization
persistent outer halo structures
bounded large-scale organization

Confirmed across multiple seeds and parameter regions.

2. MULTI-HALO COEXISTENCE

Demonstrated:

two-halo coexistence
triple-halo coexistence
interaction boundaries
spacing thresholds

Observed:

stable coexistence above critical spacing thresholds.

3. MOVING-CENTER TRACKING

Structured halos remained organized around moving centers when coupling strength was tuned appropriately.

4. ORBIT-LIKE MOTION

Observed:

stable orbit-style motion
repeatable orbital configurations
moderate mass-asymmetry survival

5. COLLISION REORGANIZATION

Direct and glancing collisions demonstrated:

- structure survival
- post-impact reorganization
- retained strong centers
- persistent coherent motion

Visual tests showed collision resilience rather than immediate random dispersal.

6. LARGE-N VISUAL SCALING

Visual organization remained stable at:

N = 250
N = 350
N = 500

Observed:

spiral structures
coherent motion
halo persistence

visual re-merger behavior

7. COMPRESSION / REBOUND MODES

Strong-center compression experiments produced:

inward spiral behavior
dense central convergence
rebound/re-expansion cycles
breathing-mode oscillatory dynamics

Compressed states frequently transitioned into reorganized
re-expansion phases while preserving particle/color identity.

8. PRECESSION RESPONSE TESTS

A baseline orbit configuration produced near-zero precession.

Modified interaction corrections then produced controlled,
scalable perihelion-style precession responses.

Observed trend:

larger correction -> larger precession
smaller correction -> smaller precession

9. PATH-CURVATURE / LENSING BEHAVIOR

Photon-like trajectory tests demonstrated:

distance-dependent path bending
mass-dependent trajectory curvature
multi-ray lensing-style behavior

Observed trend:

closer rays -> stronger bending
larger mass -> stronger curvature

10. BINARY INSPIRAL DYNAMICS

Stable binary systems combined with weak deterministic energy-loss terms produced:

gradual inward spiral
compact post-inspiral states
oscillatory rebound behavior

Observed trend:

larger decay term -> faster inspiral

11. COORDINATED SCALE TESTING

Simple geometric scaling alone failed to preserve organization.

However, coordinated scaling of:

spacing
interaction range
softening
timestep
velocity scale

produced partially preserved halo-ratio behavior.

This suggests possible approximate dimensionless scaling behavior within limited operating regions.

CURRENT SAFE INTERPRETATION

The framework currently demonstrates that tuned balancing interactions in many-body systems can generate rich gravity-like organizational behavior under coordinated parameter regimes.

The project does NOT currently claim:

- equivalence to gravity
- General Relativity
- astrophysical accuracy
- cosmological validity
- universal scale invariance
- experimentally verified physical correspondence
- dark matter replacement
- black-hole physics
- physical law derivation

Observed behaviors are currently interpreted as synthetic organization effects within a nonlinear many-body interaction framework.

CURRENT DEVELOPMENT INTERESTS

The project is currently seeking:

- GPU/simulation engineering collaboration
- nonlinear dynamics expertise
- visualization/simulation partnerships
- technical feedback
- computational systems discussion
- research-oriented collaboration

Particular interest areas include:

- GPU compute implementation
- large-scale parameter sweeps
- attractor mapping
- dimensional analysis
- swarm / organization systems

- many-body interaction research
- educational / visualization systems

The current codebase and simulation environment are available for reasonable collaboration discussion with researchers, technical partners, universities, simulation engineers, or GPU-focused development groups.

CURRENT POSITION

The strongest overall result is the transition from random particle motion to repeatable structured organization across multiple simulation branches.

The framework consistently demonstrated that tuned balancing rules can generate:

organization
persistence
orbital behavior
collision resilience
trajectory curvature
rebound dynamics
and large-scale visual structure

within a synthetic many-body interaction environment.

FINAL STATUS

MODE is currently stable enough for technical demonstration, benchmark visualization, simulation discussion, and collaborative development conversations.

Additional parameter regions, interaction architectures, and large-scale stability behaviors remain under active investigation.

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END DOCUMENT